Thu, Nov. 20

Exam 3

8:00 – 9:15

Arrive at least 5 min early!

Stepan Center

Format:

11 multiple choice questions (Only 10 (best) counts)

3 free response questions

Practice exams on Canvas

The exam will cover: everything from **Polar Coordinates** to **Fundamental Thm of line integrals**

Lectures 23 – 32

See also Review 6

Calculators: **NOT** allowed

Cylindrical (r, θ, z)	Rectangular (x, y, z)	Spherical (ρ, θ, φ)
r	?	?
?	?	$(8, \pi/4, \pi/6)$
?	$y = z + 2$	?
?	$x^2 + y^2 + z^2 = 4$	?
?	$x^2 + y^2 = 4$	?

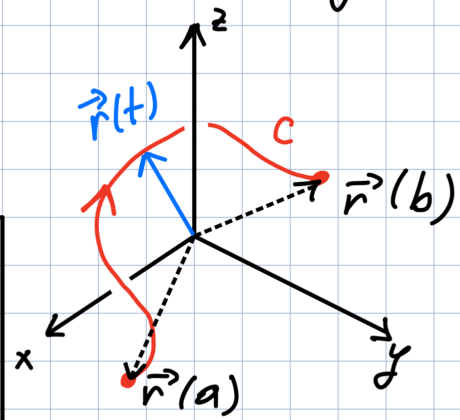
$\iiint_E F(x, y, z)\, dV =$? (in spherical coordinates)

$\iiint_E F(x, y, z)\, dV =$? (in cylindrical coordinates)

True or false?

$\vec{F} = \langle P, Q, R \rangle$ - vector field

$C: \ \vec{r}(t) = \langle x(t), y(t), z(t) \rangle$



$f(x,y,z)$ - function

1) $\displaystyle\int_C \vec{F} \cdot d\vec{r} = \int_C P\,dx + Q\,dy + R\,dz$?

2) $\displaystyle\int_C \vec{F} \cdot d\vec{r} = \vec{F}(\vec{r}(b)) - \vec{F}(\vec{r}(a))$?

3) $\displaystyle\int_C f(x,y,z)\,ds = \int_a^b f(x(t), y(t), z(t))\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}\,dt$?

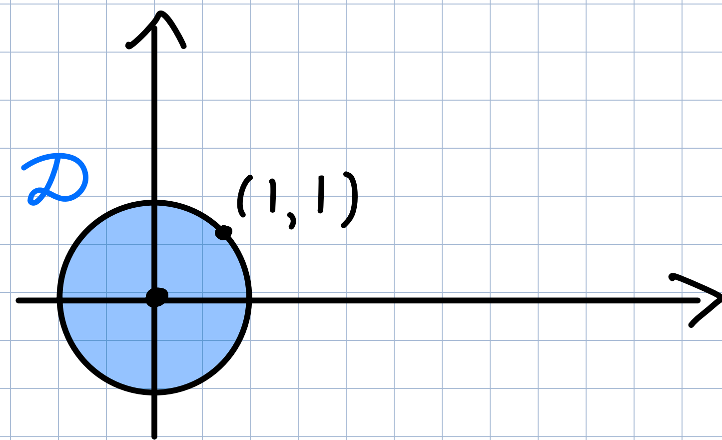
4) $\displaystyle\int_C f(x,y,z)\,dx + f(x,y,z)\,dy + f(x,y,z)\,dz = \int f(x,y,z)\,ds$?

5) $\displaystyle\int_C f(x,y,z) \cdot d\vec{r} = \int_C f_x(x,y,z)\,dx + f_y(x,y,z)\,dy + f_z(x,y,z)\,dz$?

6) $\displaystyle\int_C \nabla\vec{F} \cdot d\vec{r} = \vec{F}(\vec{r}(b)) - \vec{F}(\vec{r}(a))$?

1) Rewrite in polar coordinates:

$$\iint_{D} \frac{x}{y} + 4x^2 \, dA$$



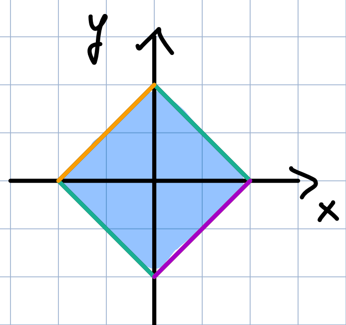
2) $\begin{cases} u = 3y - 2x \\ v = 3x - 2y \end{cases}$

a) Find the Jacobian $\dfrac{\partial(x,y)}{\partial(u,v)}$

b) $\displaystyle\iint_{R} x + y \, dA = \iint_{S} \qquad\qquad \, du\, dv$

3) $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$ $\qquad \dfrac{\partial(x,y)}{\partial(r,\theta)} = \, ?$

4)



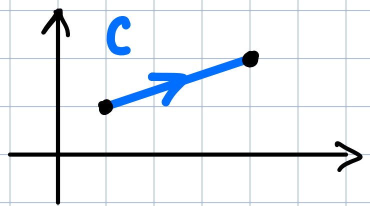
$\begin{cases} u = x + y \\ v = x^2 \end{cases}$

Sketch the region in uv-plane

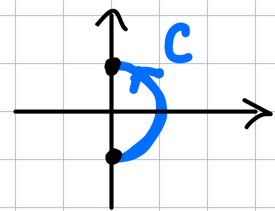
5) $\vec{F} = \langle \sin x,\ 2x + y \rangle$ and $f(x,y) = e^{xy} - y^2$

$\int_C \vec{F} \cdot d\vec{r} = \int_{\square}^{\square} \square \, dt$ $\qquad$ $\int_C f(x,y)\, ds = \int_{\square}^{\square} \square \, dt$ $\qquad$ $\int_C f(x,y)\, dy = \int_{\square}^{\square} \square \, dt$

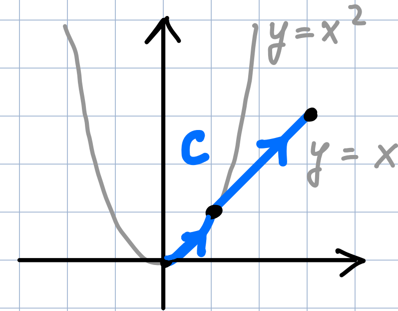
a)



b)



c)



6) Find the potential function of the vector field if it exists.

a) $\vec{F} = \langle 2x + 3,\ y - 1 \rangle$

b) $\vec{F} = \langle e^x - 2y,\ -2x \rangle$

c) $\vec{F} = \langle \sin x + e^y,\ \cos y \rangle$